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The effect of a systematic change in surface roughness skewness on turbulence and drag

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ABSTRACT

Previous work by the authors (Flack and Schultz, 2010) has identified the root-mean-square roughness height, k_{rms}, and the skewness, Sk, of the surface elevation distribution as important parameters in scaling the skin-friction drag on rough surfaces. In this study, three surfaces are tested in turbulent boundary layer flow at a friction Reynolds number, $Re_\tau = 1600$–2200. All the surfaces have similar root-mean-square roughness height, while the skewness is varied. Measurements are presented using both two-component LDV and PIV. The results show the anticipated trend of increasing skin-friction drag with increasing skewness. The largest increase in drag occurs going from negative skewness to zero skewness with a more modest increase going from zero to positive skewness. Some differences in the mean velocity and Reynolds stress profiles are observed for the three surfaces. However, these differences are confined to a region close to the rough surface, and the mean velocity and Reynolds stress profiles collapse away from the wall when scaled in outer variables. The turbulence structure as documented through two-point spatial correlations of velocity is also observed to be very similar over the three surfaces. These results support Townsend's (1976) concept of outer-layer similarity that the wall boundary condition exerts no direct influence on the turbulence structure away from the wall except in setting the velocity and length scales for the outer layer.

1. Introduction

Predicting the skin-friction drag of a generic roughness based on its surface topography is an important, yet unresolved objective in fluid mechanics research. Nikuradse (1933) carried out seminal research into roughness effects in wall-bounded turbulent flows. His study employed sieved, close-packed sand grains attached to the walls of a pipe. The resulting friction factor in the pipe was related to size of the sand grains used. This led to the widespread adoption of the equivalent sand roughness height, k_s, to characterize the frictional losses of all rough surfaces. Based on the research of Nikuradse (1933) and Colebrook (1939), Moody (1944) developed a practical engineering tool to predict the frictional losses in pipes. This is the well-known Moody diagram. It is difficult to overstate the impact the Moody diagram has had. For example, it is still prominently featured in undergraduate texts and used by practicing engineers some 75 years after its development. However, use of the Moody diagram requires specifying an equivalent roughness height or k_s for the surface in question, and k_s is only a physical roughness scale for uniform, close-packed sand. For all other surfaces, it is a hydraulic length scale that must be determined via numerical or physical experiment. Therefore, the ability to link the hydraulic length scale of a surface to its physical topography is not straightforward and remains the 'holy grail' of hydraulic research.

The past decade has seen tremendous interest, and progress, in better linking the physical characteristics of rough surfaces to their hydraulic impact. This has occurred on both the computational and experimental fronts. Numerical studies have been aided by both an increase in computational speed (Jimenez and Moser, 2007) and the better ability to model arbitrary wall boundary conditions (i.e. immersed boundary methods; Mittal and Iaccarino, 2005). These improvements have allowed studies at higher Reynolds numbers and over more complex roughness topography (e.g. Napoli et al., 2008; Yuan and Piomelli, 2011; Chan et al., 2015; Forooghi et al., 2017; Thakkar et al., 2017, Jelly and Busse, 2018). Experimental efforts (e.g. Flack and Schultz, 2010; Mejia-Alvarez and Christensen, 2010; Barros et al., 2018, Flack et al., 2020) have been assisted by techniques such as rapid prototyping, which has increased in both speed and fidelity. The upshot of this is that both computational and experimental studies can now more systematically

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explore the parameter space of surface statistics rather than simply testing a range of unrelated surface roughness conditions.

Based on a review of the literature, Flack and Schultz (2010) concluded that while a range of topographical parameters of rough surfaces correlate significantly with the hydraulic length scale, the root-mean-square roughness height, k_{rms}, and the skewness, Sk, of the surface elevation distribution show the strongest correlation with k_s. The statistical parameter of skewness physically represents whether a rough surface is dominated by peaks ($Sk > 0$) or pits ($Sk < 0$). The influence of k_{rms} and Sk on drag was critically evaluated experimentally in a fully developed turbulent channel flow by Flack et al. (2020). Results showed that surfaces with positive skewness display much larger drag penalties as compared to negatively skewed surfaces. For the cases tested, skewness also had a larger influence on drag than *rms* roughness height.

Recent direct numerical simulations (DNS) of fully developed turbulent channel flow have also investigated the influence of skewness. Thakkar et al. (2017) presented results for 17 industrially relevant irregular rough surfaces scaled to the same roughness height. The surface parameters that best correlated with drag in the transitionally rough regime were k_{rms}, Sk, frontal solidity and the streamwise correlation length. Skewness was also an important parameter for the magnitude of the peak turbulent kinetic energy. Similarly, Forooghi et al. (2017) investigated 38 surfaces with randomly distributed roughness elements of random size and prescribed shape. Skewness was identified at an important parameter in a predictive correlation for drag along with other surface statistics including effective slope, maximum peak-to-trough roughness height, and a height distribution parameter. Skewness was the key surface parameter probed by Jelly and Busse (2018), who investigated flow over a rough surface with a Gaussian distribution of surface elevations ($Sk = 0$) as well as this surface decomposed into 'pits-only' ($Sk < 0$) and 'peaks-only' ($Sk > 0$) components. The drag penalty for the negatively skewed surface was significantly lower than the other two surfaces that had peaks. This may be due to fluid filling in the surface depressions resulting in flow skimming over the pits. Jelly and Busse (2018) also investigated turbulent stresses, reporting that the the contributions of dispersive and Reynolds shear stresses show a compensating effect above the Gaussian and peaks-only surfaces. An additive effect was noted above the pits-only surface.

These results indicate that the near-wall turbulence is significantly influenced by peaked or pitted roughness morphology. The question remains if the signature of the rough surface extends further into the boundary layer. If the effect is limited to the near wall region, the surfaces would follow Townsend's (1976) outer-layer similarity hypothesis, which states that details of the wall boundary condition have no direct influence on the turbulence structure away from the wall except in setting the velocity and length scale for the outer flow. Numerous studies have critically evaluated outer layer similarity for a range of surfaces, showing that modifications to the mean flow and turbulence statistics (i.e. Flack et al., 2005; Wu and Christensen, 2007; Amir and Castro, 2011), as well as turbulent coherent structures (Volino et al., 2007; Wu and Christensen, 2010) are limited to a near-wall roughness sublayer. The effect of skewness on turbulence statistics and structures has not been isolated. This is especially interesting since the roughness skewness has such a profound effect on drag.

In the present study three surfaces with fixed k_{rms} but varying Sk (positive, zero and negative) are experimentally tested in a turbulent boundary layer flow. Detailed boundary layer profiles and velocity fields are obtained to investigate the effect of varying Sk on the mean flow, turbulence statistics and turbulence structure.

2. Experimental facilities and methods

The experiments were conducted in a boundary layer water tunnel facility. The test section is 2 m long, 0.2 m wide, and nominally 0.1 m high. The lower wall is a flat plate which serves as the test wall. The upper wall is adjustable and was set for a zero streamwise pressure gradient with a freestream velocity of ~1.0 ms^{-1} for all cases. The upper wall and sidewalls provided optical access. The boundary layer was tripped near the leading edge with a 0.8 mm diameter wire, ensuring a turbulent boundary layer. Flow was supplied to the test section from a 4000 L cylindrical tank. Water was drawn from the tank to two variable speed pumps operating in parallel and then sent to a flow conditioning section consisting of a diffuser containing perforated plates, a honeycomb, three screens and a three-dimensional contraction. The test section followed the contraction. The free stream turbulence level was <0.5%. Water exited the test section through a perforated plate emptying into the cylindrical tank. The test fluid was filtered and deaerated water. A chiller was used to keep the water temperature constant to within 1 °C during all tests. The experimental facility is shown in Fig. 1.

Boundary-layer velocity measurements were taken 127 cm downstream from the leading edge of the roughness and 150 cm downstream of the trip. Measurements were obtained with a TSI FSA3500 two-component laser-Doppler velocimeter (LDV). The LDV consists of a four-beam fiber optic probe that collects data in backscatter mode. A custom-designed beam displacer was added to the probe to shift one of the four beams, resulting in three co-planar beams that can be aligned parallel to the wall. Additionally, a 2.6:1 beam expander was located at the exit of the probe to reduce the size of the measurement volume. The probe volume diameter, d, of the LDV is 45 μm which is $d^+ < 2.6$ for the measurements reported herein. Velocity profiles consisted of ~50 measurement locations at varying wall-normal distances. Data were acquired at a given sampling location for 540 s or until 120,000 velocity realizations were made. The later results in a shorter sampling time. Due to the variation in data rate with distance from the wall, the number of realizations varied from 30,000 to 120000. The data were collected in coincidence mode. The flow was seeded with 2 μm silver-coated glass spherical particles. Further details of the experimental facility and LDV system can be found in Volino et al. (2007). Uncertainty estimates in the LDV measurements were determined by combining bias with the precision uncertainties, as detailed in Schultz and Flack (2007). The overall uncertainty in the mean velocity is ± 0.5% and for the turbulence quantities $\overline{u'^2}$, $\overline{v'^2}$, and $\overline{u'v'}$, it is ± 3%, ± 4%, and ± 6%, respectively.

The experimental conditions are listed on Table 1. The friction velocity, $U_\tau = \sqrt{\tau_w/\rho}$, where τ_w is the wall shear stress and ρ is the fluid density, was determined using a modified Clauser chart method of Perry and Li (1990), as described in Schultz and Flack, (2003). This iterative procedure using the velocity profile in the log-law region determines both the wall shear stress and the wall datum offset, or location in the roughness where $y = 0$. U_τ was verified via the total stress method (Schultz and Flack, 2007) using the plateau in the total stress (i.e. Reynolds shear plus viscous shear in the log-layer). The values of U_τ agreed within ± 2% in all cases. The uncertainty in U_τ for the Clauser chart method is ± 4% and ± 5% for the total stress method.

Velocity field data were acquired using particle image velocimetry

Fig. 1. Boundary layer water tunnel facility.

Table 1
Experimental conditions.

Surface	U_τ (ms^{-1})	ΔU^+	C_f ($\times 10^3$)	δ (mm)	Re_τ	k_s^+	$k_s/\bar{k_t}$	$\bar{k_t}$ /δ
Negative Skewness	0.0482	5.49	4.65	32.2	1610	44.5	0.26	0.107
Zero Skewness	0.0530	7.93	5.51	35.1	1920	114	0.64	0.091
Positive Skewness	0.0556	8.92	5.94	38.4	2200	167	0.84	0.090

(PIV) at the same streamwise location as the LDV profiles. For each measurement plane, 1000 image pairs were acquired using a CCD camera with a 3320 × 2496 pixel array. Image processing was done with TSI Insight 4G software. Velocity vectors were obtained using 32 pixel square windows with 50% overlap. A streamwise-wall normal plane was acquired at the spanwise centerline of the test section, and at $y/\delta = 0.15$ in the streamwise-spanwise plane.

Two-point spatial correlations were obtained for each measurement plane. In the (xy) plane the correlation is defined at the wall normal position y_{ref} as

$$R_{AB}(y_{ref}) = \frac{\overline{A(x, y_{ref})B(x + \Delta x, y_{ref} + \Delta y)}}{\sigma_A(y_{ref})\sigma_B(y_{ref} + \Delta y)} \tag{1}$$

where A and B are the quantities of interest at two locations separated in the streamwise and wall normal directions by Δx and Δy, and σ_A and σ_B are the standard deviations of A and B at y_{ref} and $y_{\text{ref}} + \Delta y$, respectively. At every y_{ref}, the overbar indicates the correlations were averaged among location pairs with the same Δx and Δy, and then time averaged over the 1000 vector fields. Correlations of u, v, and the cross-correlations of u and v were considered. For the xz-planes, the correlation is defined as

$$R_{AB}(y) = \frac{\overline{A(x, z)B(x + \Delta x, z + \Delta z)}}{\sigma_A(y)\sigma_B(y)} \tag{2}$$

where A and B are the quantities of interest at two locations separated in the streamwise and spanwise directions by Δx and Δz, and σ_A and σ_B are the standard deviations of A and B based on data in the full plane for the 1000 vector fields. The correlations were averaged among all locations with the same Δx and Δz, and then time averaged.

The rough surfaces were generated mathematically so the surface statistics could be systematically altered. The surfaces were generated in MATLAB using a circular Fast Fourier Transform (FFT) with a random set of independent phase angles, distributed between 0 and 2π, with a Gaussian power spectral density (PSD), in the form of $E(k) = lh^2/(2\sqrt{\pi})e^{-k^2l^2/4}$, where k is the wavenumber, and l and h are parameters that control the shape of the spectrum (l sets the length-scale of the roughness elements and h the roughness *rms*). The random phase was generated using the Pearson system random numbers (Johnson et al. 1994), where both skewness (*Sk*) and flatness (*Fl*) can be set as input parameters. These values were adjusted until the desired *Sk* and *Fl* were achieved after the Gaussian power spectrum was imposed. Even though a Gaussian power spectrum was chosen, the skewed surfaces possess a non-Gaussian probability density function. The skewness of the pdf was varied with negative, positive and zero skewness over the range $-1.0 < Sk < +1.0$. The roughness heights (mean, k_a, root-mean-square, k_{rms}, average peak-to-trough, $\bar{k_t}$) were all kept approximately constant. The MATLAB generated surfaces were reproduced using a high-resolution 3D printer (Objet30 Pro) with a lateral resolution of 34 µm and a vertical resolution of 16 µm. Images of the three test surfaces are shown in Fig. 2. Three different plates were produced for each roughness case with dimensions of 28.0 cm by 20.3 cm in the x and z directions, respectively. The test surface comprised six total plates arranged in a random manner to avoid any repeating roughness features that could produce secondary flows.

Table 2 lists the statistics of the surfaces for the entire sampled

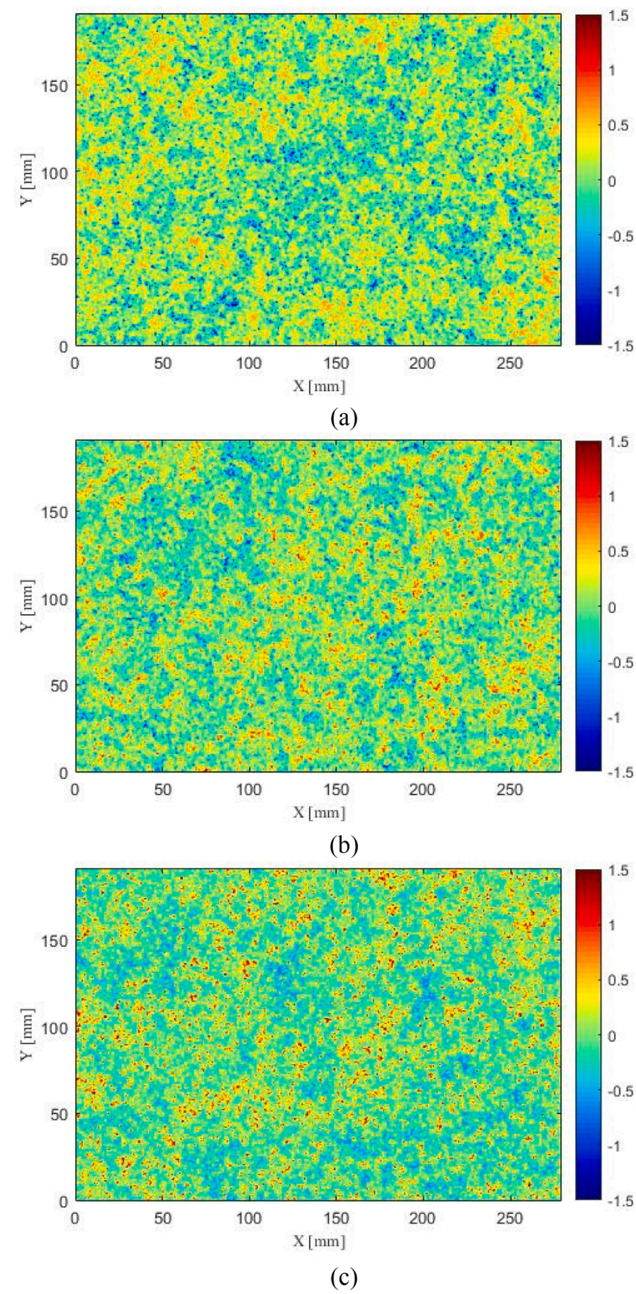

Fig. 2. Generated test surfaces: (a) $Sk = -1$ case; (b) $Sk = 0$ case; (c) $Sk = +1$ case. Colors indicate roughness height in mm.

region, where h' is the variation in roughness elevation about the mean and A_t is the total plan area. These include the average roughness height, $k_a = \frac{1}{A_t}\int |h'| dA$, the *rms* roughness height, $k_{rms} = \sqrt{\frac{1}{A_t}\int |h'^2| dA}$, the average peak-to-trough roughness height for the N = 5 largest

Table 2
Rough surface statistics.

	Negative Skewness	Zero Skewness	Positive Skewness
Mean amplitude, k_a (μm)	276	280	277
Root-mean-square height, k_{rms} (μm)	350	350	350
Average peak-to-trough height, $\bar{k}_t$ (μm)	3449	3244	3474
Skewness, *Sk*	−0.97	0.00	+0.98
Flatness, *Fl*	4.17	3.00	4.18
Effective slope, *ES*	0.40	0.44	0.40
Correlation length, *R* (μm)	1090	1060	1090

differences, $\bar{k}_t = \frac{1}{N}\sum_{i=1}^{N}\left(h'_{max_i} - h'_{min_i}\right)$, skewness, $Sk = \left(\frac{1}{A_t}\int h'^3\, dA\right) \Big/ k_{rms}^3$, flatness, $Fl = \left(\frac{1}{A_t}\int h'^4\, dA\right) \Big/ k_{rms}^4$, and the effective slope, $ES = \frac{1}{A_t}\int \left|\frac{\partial h'}{\partial x}\right| dA$, (Napoli et al. 2008). The correlation length, R, of the surface height, h', is the value of Δx where the function $C(\Delta x) = h'(x,z)h'(x+\Delta x,z)/k_{rms}^2$ attains a value of 0.25.

3. Results and discussion

The mean velocity profiles are presented in Figs. 3 and 4. Fig. 3 shows that all the rough-wall profiles display a downward shift in the log law region, or roughness function (Hama 1954), ΔU^+, compared to the smooth-wall results of Sillero et al. (2014) at a similar Reynolds number ($Re_\tau = U_\tau\, \delta/\nu \sim 2000$). The equivalent sandgrain roughness height, k_s, is the roughness height that produces the same roughness function as the uniform sandgrain of Nikuradse in the fully rough regime. Using the roughness function, ΔU^+, k_s can be determined from $\Delta U^+ = \frac{1}{\kappa}\ln k_s^+ + A - B$ where $\kappa = 0.384$ is the von Kármán constant, $B = 4.1$ is the log-law intercept for a smooth wall, and $A = 8.5$ is the intercept for a uniform sandgrain surface. The minimum k_s^+ for fully-rough behaviour has proven to be roughness dependent, as discussed in Flack and Schultz (2010). While $k_s^+ = 44.8$ for the negatively skewed case is less than $k_s^+ = 70$, the fully rough threshold for Nikuradse (1933) sandgrain, a geometrically similar negatively skewed roughness with a Gaussian PSD distribution of scales that was tested over a wide range of roughness Reynolds numbers indicated fully rough behaviour for $k_s^+ \sim 40$ (Flack, et al. 2020). The roughness function, ΔU^+, skin friction coefficient, $C_f = \frac{1}{2}\tau_w/\rho U_e^2$, where U_e is the freestream velocity, and roughness Reynolds number, $k_s^+ = U_\tau\, k_s/\nu$, are listed on Table 1.

It can be seen that the primary effect of going from a negatively skewed roughness to a positively skewed one is an increase in the skin-friction coefficient, C_f, the boundary layer thickness, δ, and the equivalent sand roughness height, k_s. The largest increase in these quantities is observed to occur when moving from negative to zero skewness, while the increase is more modest going from zero to positive skewness. For example, the skin-friction coefficient, C_f, increases by nearly 20% between the negatively skewed and the zero skewed roughness, while the increase observed between the positively skewed and zero skewed roughness is <10%. This is also illustrated by the ratio of the hydraulic roughness height to the physical roughness height, $k_s/\bar{k}_t$, also listed on Table 1. The ratio $k_s/\bar{k}_t$ increases by a factor of nearly 2.5 times between the negatively skewed and the zero skewed roughness, while the increase observed between the zero skewed and positively skewed roughness is about 1.3 times. An increase in skin-friction and related quantities with increasing skewness is an expected result. The relative magnitude of the increases seen here is quite similar to that recently observed in the numerical study of Jelly and Busse (2018). They investigated roughness topographies with only peaks, only pits, and both peaks and pits and found that the largest increase in skin-friction occurs going from the pits-only surface to one with both peaks and pits. Flack et al. (2020) also observed a marked increase in skin-friction with increased skewness for a similar stochastic roughness. This related study observed a significant increase in skin-friction for the positive skewed surfaces, whereas there was a relatively small difference in the skin-friction coefficient for the negatively skewed surfaces compared to the surface with zero skewness. Similar to Jelly and Busse (2018), flow skimming of surface depressions may play a larger role for the current surface texture. While skewness is an important parameter for predicting surface drag, other surface features are obviously also important.

Fig. 4 shows the mean profiles in outer variables. Here it is seen that all of the profiles collapse in velocity-defect form for $y/\delta > 0.04$ supporting the notion of outer-layer similarity in the mean flow. A mean flow roughness sublayer of $y/\delta \approx 0.04$ corresponds to approximately 0.4 $\bar{k}_t$, 4 k_{rms} or 1.4 k_s, 0.7 k_s or 0.5 k_s, for negative, zero or positive skewness, respectively. The majority of the roughness literature supports outer-layer similarity in the mean velocity profile. Conventional thought is that similarity can be expected to hold provided the relative roughness height ($\bar{k}_t/\delta$) is not too large. In the present work, the $\bar{k}_t/\delta$ is $\sim$ 10%. Connelly et al. (2006) observed outer-layer similarity in the mean flow for surfaces with a relative roughness height that was comparable to the present study. Castro (2007) asserted that mean flow universality holds for surfaces with $\bar{k}_t/\delta$ up to ~20% or greater. Recent results seem to indicate that a large relative roughness height might not be a reliable indicator for the breakdown of mean flow similarity. Instead, secondary flows can be generated on roughness with spanwise and/or streamwise periodicity with wavelengths comparable to the boundary layer thickness, even in cases of rather small relative roughness height. Spanwise

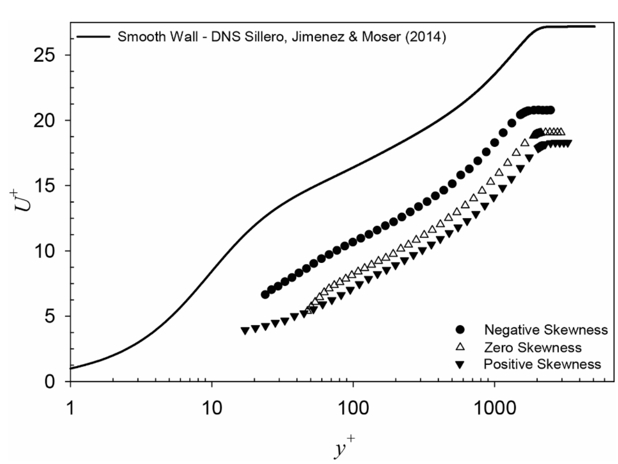

Fig. 3. Mean velocity profiles in inner variables.

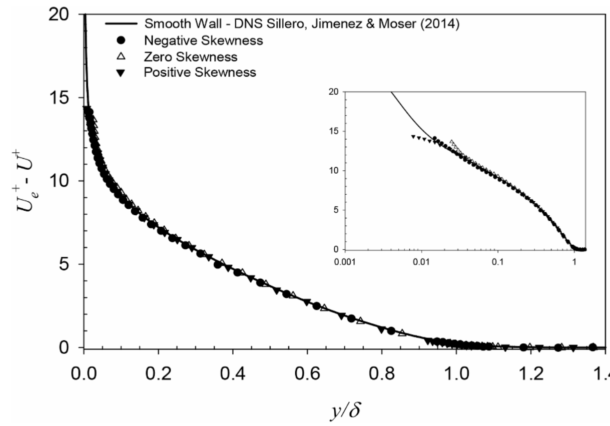

Fig. 4. Mean velocity profiles in velocity-defect form.

gradients in Reynolds shear stress can give rise to large-scale roll modes that lead to significant changes in the mean flow well into the outer layer and cause a breakdown in mean flow similarity (Barros and Christensen, 2014). Secondary flows are not expected for the current roughness since the correlation length (listed on Table 2) is <4% of the boundary layer thickness.

The Reynolds stress profiles are presented in Figs. 5–7. The streamwise Reynolds normal stress ($\overline{u'^2}$) profiles (Fig. 5) indicate that the primary effect of the roughness is to suppress the near-wall peak in $\overline{u'^2}$ that occurs on the smooth wall. This observation is in agreement with the results of Ligrani and Moffat (1986). Their results showed a gradual reduction in the near-wall peak in $\overline{u'^2}$ with increasing roughness Reynolds number until the fully-rough regime is reached where a near total destruction in the peak is observed. The roughness Reynolds number range ($45 \le k_s^+ \le 167$) and the destruction of the near-wall peak in $\overline{u'^2}$ indicates that the present cases are in the fully-rough flow regime or nearly so. Differences in the profiles of $\overline{u'^2}$ for the rough walls are limited to $y/\delta < 0.1$. Farther from the wall, the profiles collapse and agree within experimental uncertainty with the smooth-wall profile of Sillero et al. (2014) at a similar Reynolds number.

The wall-normal Reynolds normal stress ($\overline{v'^2}$) profiles for the rough walls (Fig. 6) show collapse across most of the boundary layer. Very close to the roughness ($y/\delta < 0.05$), $\overline{v'^2}$ appears to be enhanced for the positively skewed roughness. The present rough-wall results also agree quite well with the smooth-wall experimental results of DeGraaff and Eaton (2000) taken at a similar Reynolds number. The present rough-wall results and the smooth-wall results of DeGraaff and Eaton, acquired via LDV, lie systematically above the Sillero et al. (2014) results as one gets farther from the wall and the freestream is approached. Since the freestream turbulence level in both the present study and the work of DeGraaff and Eaton is quite low, the fact that the wall-normal Reynolds normal stress does not go to zero in the freestream cannot be physical. The maximum boundary layer thickness to tunnel half-height also indicates that there is a core flow in the boundary layer tunnel. This systematic overestimation of turbulence in the freestream with LDV was previously noted by DeGraaff (1999). DeGraaff hypothesized the error to be due to a slight fringe aberration resulting from the small probe volume. He showed compelling evidence that this aberration leads to overestimation of the turbulence when the turbulence intensity is low (i.e. in the freestream), but the effect becomes negligible when the actual turbulence intensity is increased (i.e. in the boundary layer). Therefore, this effect is expected to influence only measurements near the outer edge of the boundary layer and in the freestream and not those taken deeper in the boundary layer.

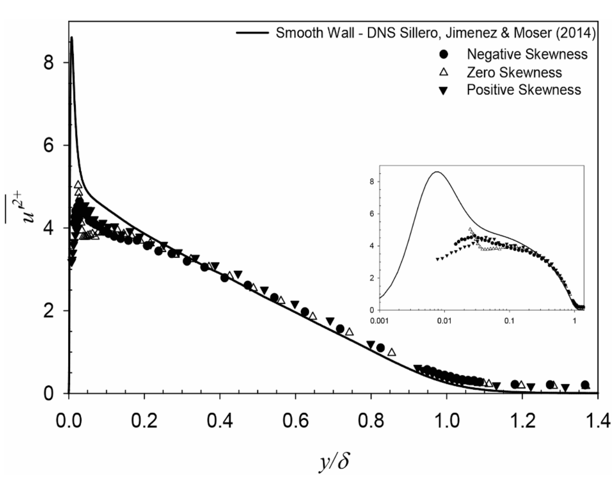

Fig. 5. Streamwise Reynolds normal stress in outer variables.

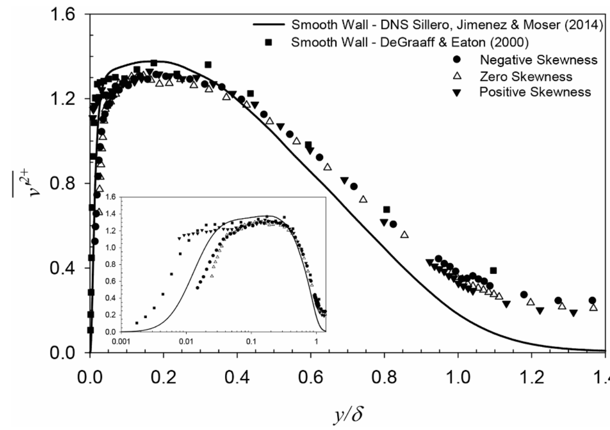

Fig. 6. Wall-normal Reynolds normal stress in outer variables.

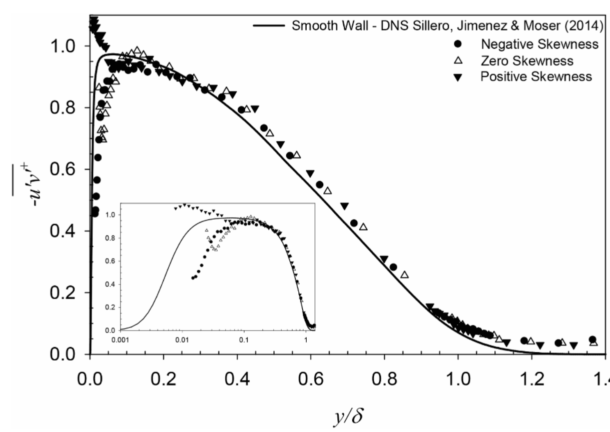

Fig. 7. Reynolds shear stress in outer variables.

The Reynolds shear stress profiles for the rough walls (Fig. 7) also collapse across most of the boundary layer. Very near the roughness ($y/\delta < 0.05$), the Reynolds shear stress appears to be enhanced for the positively skewed roughness as compared to the other cases. This may be a result of flow separation and vortex shedding from the sharp features of the positively skewed surface. The present rough-wall results also agree within experimental uncertainty with the smooth-wall DNS results of Sillero et al. (2014) throughout most of the boundary layer. Again, this supports outer layer similarity outside of a near-wall roughness sublayer of approximately $1\bar{k}_t$, $8k_{rms}$ or $3k_s$ for the Reynolds stresses of the three surfaces.

Fig. 8 shows results from PIV taken in the streamwise-wall-normal plane at $y/\delta = 0.15$. Contours of the two-point correlations, R_{uu} (Fig. 8a–c), R_{vv} (Fig. 8f–h), and R_{-uv} (Fig. 8k–m) are presented for the negative, zero, and positive skewed rough surfaces. The contours show that the overall shape of the correlations for all of the rough surfaces is very similar. The correlations of R_{uu} are elongated in the streamwise direction and are inclined at an angle of ~11–13° to the wall. The angle is comparable to that observed by Wu and Christensen (2010) (~10°) for both smooth and rough walls at $y/\delta = 0.15$. This spatial signature is indicative of the hairpin packet vortical structure presented by Adrian et al. (2000). R_{vv} is much less elongated in the streamwise and wall normal direction. R_{-uv} shows some streamwise elongation, which is largest in extent for the peak-dominated roughness ($Sk = +1$). The near-wall enhancement in Reynolds shear stress for this roughness has some remnants in the spatial signature of R_{-uv} at $y/\delta = 0.15$. Fig. 8 also shows streamwise and wall normal slices of the R_{uu}, R_{vv}, and R_{-uv} correlations, passing through the self-correlation peaks. It can be seen that the correlations not only look similar but are also in excellent quantitative

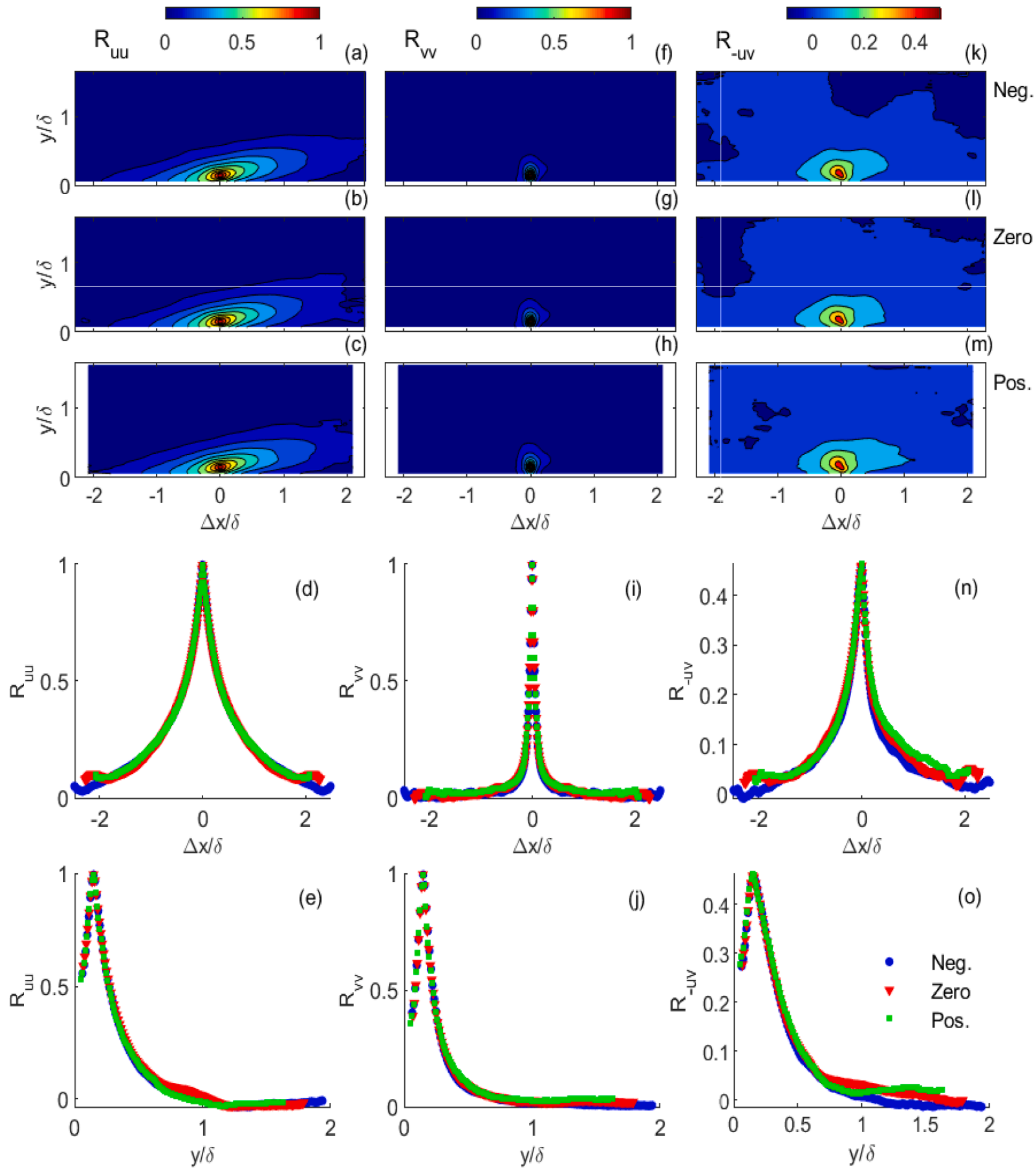

Fig. 8. Contours of two-point correlations, R_{uu} (a–c), R_{vv}, (f–h), and $R_{\text{-}uv}$ (k–m) in the streamwise-wall-normal plane centered at $y/\delta = 0.15$ for $Sk = -1$ (a, f, k), $Sk = 0$ (b, g, l), and $Sk = +1$ (c, h, m) cases. Also shown are correlation cuts taken through the R_{uu} (d, e), R_{vv} (i, j) and $R_{\text{-}uv}$ (o, p) correlation peaks in streamwise (d, i, n) and spanwise (e, j, o) direction.

agreement, except for small differences in $R_{\text{-}uv}$ in the positive streamwise direction.

Fig. 9 shows results from PIV taken in the streamwise-spanwise plane at $y/\delta = 0.15$. Contours of the two-point correlations, R_{uu} (Fig. 9a–c), R_{ww} (Fig. 9f–h), and R_{uw} (Fig. 9k–m) are presented for the negative, zero, and positive skewed rough surfaces. The overall shape of the correlation maps for the three rough surfaces again look to be quite similar. As expected, R_{uu} and R_{ww} are symmetric in the spanwise direction, whereas R_{uw} shows the signature of the legs of hairpin vortices with elongated correlations in the streamwise direction and opposite signed correlations for $\pm\,\Delta z$. Streamwise (Fig. 9d, i) and spanwise cuts (Fig. 9e, j) of the R_{uu} and R_{ww} correlations maps indicate that these correlations are in close agreement for the three rough surfaces, demonstrating strong similarity of coherent structures in spatial scales.

Similarity of spatial correlations has been seen in other rough wall studies. Wu and Christensen (2010) observed qualitative agreement of R_{uu}, R_{vv} at $y/\delta = 0.15$ for rough and smooth walls for turbine blade roughness. Quantitatively, the correlations collapsed except for slight shortening of R_{uu} in the streamwise direction. The correlations also show characteristics of the same quantities from PIV measurement over a smooth wall and mesh roughness outside the roughness sublayer at $y/\delta = 0.40$ from the study of Volino et al. (2007).

4. Summary

An experimental study was conducted in which the roughness height (k_a, k_{rms} and $\bar{k_t}$) as well as the effective slope and correlation length were held approximately constant while the skewness was systematically varied in turbulent boundary layer flow at a friction Reynolds number, $Re_\tau = 1600$–2200. While the roughness heights were fixed, the ratio of heights to boundary layer thickness (k_a/δ, k_{rms}/δ and $\bar{k_t}\,/\delta$) changed modestly due to an increase in boundary layer thickness as the skewness changed from negative to positive. The results show the expected trend of increasing skin-friction drag and other drag-related variables with increasing skewness. The largest increase is observed to occur going

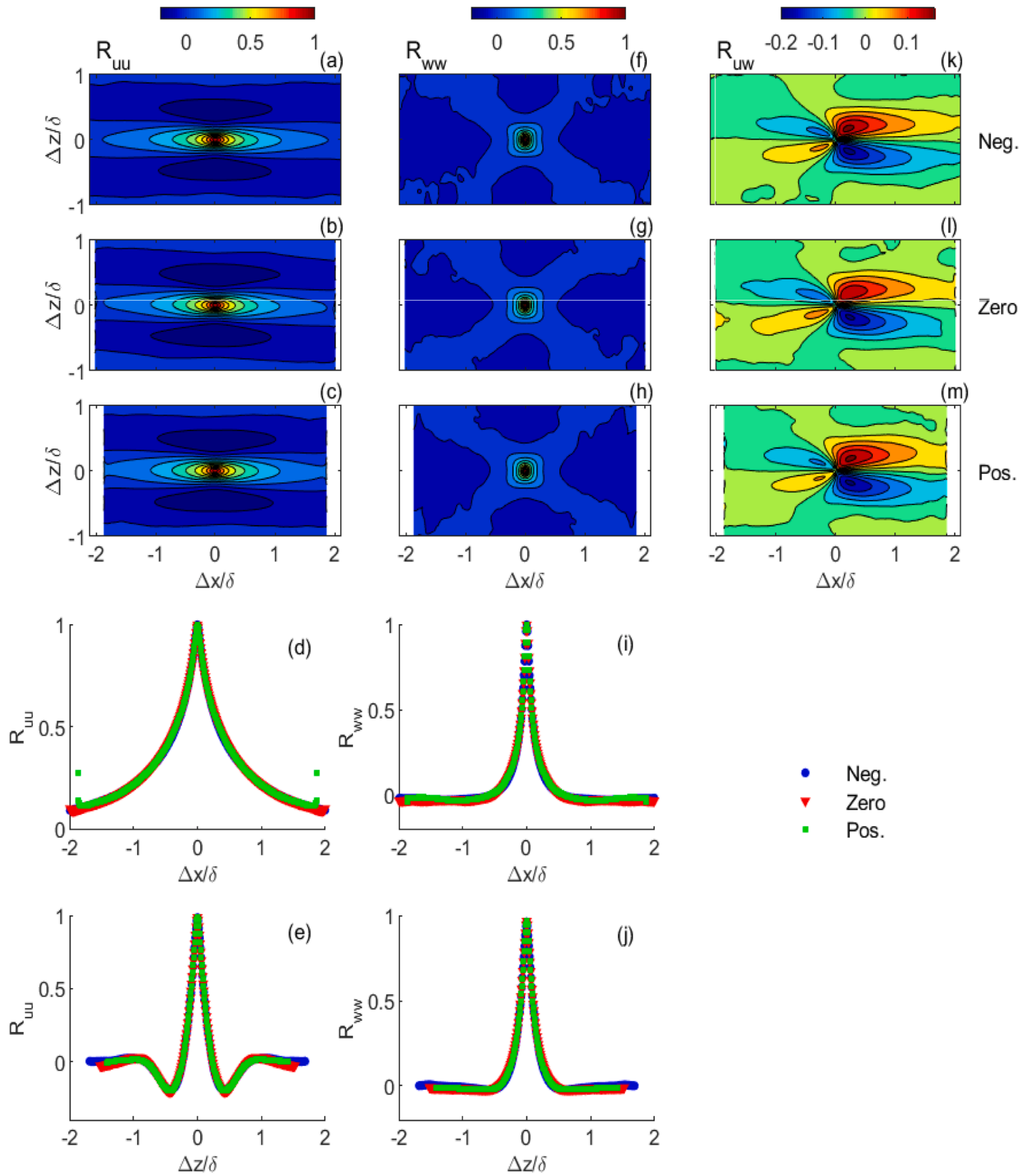

Fig. 9. Contours of two-point correlations, R_{uu} (a–c), R_{ww} (f–h) and R_{uw} (k-m), in the streamwise-spanwise plane centered at $y/\delta = 0.15$ for $Sk = -1$ (a, f, k), $Sk = 0$ (b, g, l), and $Sk = +1$ (c, h, m) cases. Also shown are correlation cuts taken through the R_{uu} (d, e) and R_{ww} (i, j) correlation peaks in streamwise (d, i) and spanwise (e, j) direction.

from negative to zero skewness with a more modest increase going from zero to positive skewness. This result agrees well with the recent numerical study of Jelly and Busse (2018) and shows the same trend with increased drag from negative to positive skewness as Flack et al. (2020). Peak dominated surfaces have significantly higher drag than pitted surfaces. The lower drag penalty for negatively skewed surfaces may be the result of the flow skimming over the surface depressions. The roughness skewness appears to be an important statistical parameter for predicting drag, especially for this close packed roughness with other statistical parameters fixed. Other features of the surface texture would impact the near wall flow for a different roughness geometry. As a minimum, skewness, combined with a surface height (likely k_a or k_{rms}) as well as a measure of the surface slope are needed for predictive correlations of drag.

Near-wall differences in the mean velocity and Reynolds stress profiles are observed for the three surfaces. This was especially evident for the Reynolds shear stress for the positively skewed surface. These differences are found to be confined to a region within one roughness height ($\bar{k_t}$) or three equivalent roughness heights (k_s) of the wall. The profiles collapse farther away from the wall when scaled in outer variables.

Two-point spatial correlations also indicate that the turbulence structure in the outer layer is not affected by the skewness of the roughness. Qualitative similarity was observed for correlations from the streamwise-wall normal and streamwise-spanwise planes. Quantitatively, all spatial correlations collapsed for the three rough surfaces except R_{uv} for the positive skewed surface which demonstrated slightly elongated correlations in the positive streamwise direction.

While a peak dominated surface has significantly higher drag than a pit dominated surface, the differences in turbulence statistics and spatial correlations are contained to a near-wall roughness sublayer, supporting Townsend's (1976) concept of outer-layer similarity. Results also support the use of wall functions in numerical simulations of rough-wall turbulent boundary layers. Adherence to outer-layer similarity was observed when the largest roughness features ($\bar{k_t}$) were approximately 10% of the boundary layer thickness. Outer layer effects may be evident if the roughness height is a significant fraction of the boundary layer thickness or in the presence of secondary flows due to periodic roughness wavelengths comparable to the boundary layer thickness.

CRediT authorship contribution statement

Karen A. Flack: Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft, Visualization, Project administration, Funding acquisition. **Michael P. Schultz:** Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft, Visualization, Funding Acquisition. **Ralph J. Volino:** Conceptualization, Methodology, Investigation, Formal analysis, Writing - original draft, Visualization, Funding Acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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